LRU PAGE REPLACEMENT

Completed

LRU Page Replacement Algorithm

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string. \
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>

int main() {
    int n, frames_count, i, j;
    int page_faults = 0, page_hits = 0;

    printf("Enter number of pages in reference string: \n");
    if (scanf("%d", &n) != 1) return 0;

    int reference_string[n];
    printf("Enter the reference string: \n");
    for (i = 0; i < n; i++) {
        scanf("%d", &reference_string[i]);
    }

    printf("Enter number of frames: \n");
    if (scanf("%d", &frames_count) != 1) return 0;
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Sample Input 1

5

1 2 3 4 5

5

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

```c
    printf("Enter number of frames: \n");
    if (scanf("%d", &frames_count) != 1) return 0;

    int frames[frames_count];
    int last_used[frames_count];

    for (i = 0; i < frames_count; i++) {
        frames[i] = -1;
        last_used[i] = 0;
    }

    for (i = 0; i < n; i++) {
        int found = 0;
        int current_page = reference_string[i];
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

5

1 2 3 4 5

5

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State
1	1	1 - - - -

Select Language : C (GCC 9.2.0)

```c
    for (i = 0; i < n; i++) {
        int found = 0;
        int current_page = reference_string[i];


        for (j = 0; j < frames_count; j++) {
            if (frames[j] == current_page) {
                found = 1;
                page_hits++;
                last_used[j] = i;
                break;
            }
        }

        if (!found) {
            int empty_slot = -1;
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

5

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State
1	1	1 - - - -
2	2	1 2 - - -

```
            }
        }

        if (!found) {
            int empty_slot = -1;

            for (j = 0; j < frames_count; j++) {
                if (frames[j] == -1) {
                    empty_slot = j;
                    break;
                }
            }

            if (empty_slot != -1) {

                frames[empty_slot] = current_page;
                last_used[empty_slot] = i;
            } else {
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State
1	1	1 - - - -
2	2	1 2 - - -
3	3	1 2 3 - -
4	4	1 2 3 4 -
5	5	1 2 3 4 5

Select Language : C (GCC 9.2.0)

```
            for (j = 0; j < frames_count; j++) {
                if (frames[j] == -1) {
                    empty_slot = j;
                    break;
                }
            }

            if (empty_slot != -1) {

                frames[empty_slot] = current_page;
                last_used[empty_slot] = i;
            } else {
                int lru_index = 0;
                int min_time = last_used[0];
                for (j = 1; j < frames_count; j++) {
                    if (last_used[j] < min_time) {
                        min_time = last_used[j];
                        lru_index = j;
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State
1	1	1 - - - -
2	2	1 2 - - -
3	3	1 2 3 - -
4	4	1 2 3 4 -
5	5	1 2 3 4 5

Select Language : C (GCC 9.2.0)

```
                    if (last_used[j] < min_time) {
                        min_time = last_used[j];
                        lru_index = j;
                    }
                }

                frames[lru_index] = current_page;
                last_used[lru_index] = i;
            }
            page_faults++;
        }
    }

    printf("\nTotal Page Faults: %d", page_faults);
    printf("\nTotal Page Hits: %d\n", page_hits);

    return 0;
}
```

Test Cases 2 | Run Sample Cases | Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1